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From Voice to Valor: An Examination of How AI Voice Modality Enhances User Social Confidence

Short Paper

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Abstract

As generative AI increasingly functions as a social partner, understanding its psychological effects on users has become important. Drawing on the Machine-Integrated Relational Adaptation (MIRA) model and Self-Expansion Theory, this study examines how AI voice modality (neutral text, machine-like voice and human-like voice) influences users' social confidence through perceived intelligence and self-expansion, with attitudes toward AI serving as a moderator. A 14-day longitudinal experiment with 125 university students was conducted in which participants interacted daily with an AI companion under different modality conditions. Preliminary results show that human-like voice significantly increases perceived intelligence; although direct effects on self-expansion and social confidence are not observed in simple group comparisons, the findings suggest indirect psychological mechanisms linking AI voice design to users' social confidence. This study extends to Human-AI interaction and social development literature and provides implications for designing AI companions that support users' psychological empowerment and social development.

Keywords: AI companion, Voice-based interaction, Perceived intelligence, Self-expansion, Social confidence, Human-AI interaction

Introduction

Generative AI has permeated the fabric of modern social life, transcending its historical function as an information processor and increasingly assuming the role of a “social partner”. This evolution is epitomized by the AI companions, sophisticated conversational agents designed not only to execute tasks but also to simulate anthropomorphic empathy, consistent personality, and sustained intimacy (Wang et al., 2024). This development marks a critical transition in which AI shifts from a functional tool to a distinct “social entity”, capable of co-creating dynamic interpersonal experiences through human-like interactions (Boyd & Markowitz, 2026).

Current research in the fields of Human-Computer Interaction (HCI) and psychology has primarily centered on the relational and supportive functions of AI companions. A large number of empirical studies have shown that AI companions can alleviate loneliness by providing the social perception of “being heard” (De Freitas et al., 2025), especially for individuals with social anxiety, as AI companions offer a non-judgmental interactive space (Wang et al., 2025). In addition, users develop a certain degree of attachment after interacting with AI companions, which helps to improve their mental health (Zhang et al., 2025). Nevertheless, existing studies have mostly focused on AI usability, satisfaction, or human-AI trust, and few have explored the consequences of human-AI interaction from the deep psychological dimension of “social confidence”, which is crucial for human social development.

Social confidence is a cognitive-emotional state characterized by subjective certainty about one's ability to navigate social situations and maintain meaningful interpersonal relationships. Drawing from prior work,

it can be understood as a socially grounded form of confidence rooted in self-understanding and expressed in individuals' perceived ability to initiate, sustain, and regulate interpersonal interaction (Hewitt & Barbalet, 2001; Hogg & Grieve, 1999). In psychological research, social confidence is often operationalized as social self-efficacy, referring to individuals' beliefs in their ability to initiate and maintain interpersonal relationships and adapt to complex social situations (Smith & Betz, 2000). As a cognitive precursor to social behavior, social confidence promotes proactive social strategies, persistence amid social setbacks, and effective regulation of interpersonal stress (Anderson & Betz, 2001). Empirically, it acts as a protective factor against psychological distress: negatively associated with social anxiety and loneliness, it positively predicts perceived social support and life satisfaction (Wei et al., 2005). Moreover, social confidence facilitates adaptive functioning across life transitions by supporting the development of meaningful social bonds, which are critical for emotional well-being and occupational success (Lee et al., 2012). Despite its importance, little is known about whether sustained interaction with AI companions can strengthen users' social confidence.

This study focuses on how the modality of AI interaction, especially highly anthropomorphic voice interaction, serves as a key factor influencing users' social confidence through a 14-day longitudinal experiment. Compared with text interaction, voice is considered to have higher immediacy and social presence (Kim et al., 2022), and can induce stronger arousal levels (Fang et al., 2025). We propose that when AI presents with a smooth human-like voice, users are more likely to perceive it as an entity with high perceived intelligence. According to the Self-Expansion Theory, when interacting with highly capable partners, individuals tend to incorporate the partners' resources, perspectives, and abilities into their own self-concept, thus generating psychological fusion (Mills et al., 2025). Applied to AI companionship, when users perceive an AI companion as highly intelligent, they may experience a sense of psychological expansion by internalizing the AI's perceived competence and support. This self-expansion process may, in turn, enhance users' evaluations of their own social capability, thereby increasing social confidence. In this way, AI companions may influence users not only by offering external support, but also by reshaping how users understand themselves.

At the same time, the effects of AI modality are unlikely to be uniform across users. We argue that attitudes toward AI play an important moderating role in this process. Attitudes toward AI reflect individuals' cognitive, affective, and behavioral predispositions toward AI technologies, and prior research suggests that more positive attitudes are associated with greater acceptance of voice assistants, robots, and other intelligent systems (Li et al., 2025). More importantly, such attitudes shape how users interpret technological cues. Users with favorable attitudes toward AI may be more likely to perceive an anthropomorphic AI companion, especially one with a human-like voice, as intelligent, agentic, and socially capable. By contrast, users with more skeptical attitudes may dismiss the same cues as superficial design features rather than meaningful signals of intelligence. Accordingly, attitudes toward AI are likely to strengthen or weaken the effect of interaction modality on perceived AI intelligence, helping explain why the same AI design can produce different psychological outcomes across users.

Accordingly, this study aims to address the following questions:

RQ1: To what extent does AI voice modality (neutral text vs. machine-like voice vs. human-like voice) influence perceived intelligence of AI, and thereby promote self-expansion and social confidence?

RQ2: How do users' existing attitudes toward AI moderate the relationship between voice modality and perceived intelligence of AI?

To systematically investigate these questions, we employ a 14-day longitudinal experiment and ground our model in the Machine-Integrated Relational Adaptation (MIRA) model (Boyd & Markowitz, 2026), which conceptualizes AI as a relational actor embedded in a dynamic social system. Within this framework, we position voice modality as the antecedent, perceived intelligence and self-expansion as the key mechanisms, social confidence as the outcome, and attitudes toward AI as the moderator. Through this, we investigated whether AI interaction modalities have an impact on users' self-perceived social confidence and revealed the influencing paths. This work is expected to contribute to the IS literature by extending research on AI companions beyond adoption- and support-oriented outcomes to examine social confidence as a novel and practically important consequence of interacting with AI companions, while also offering important design implications for voice interfaces.

Literature Review and Theoretical Foundation

AI Companion

AI companions are sophisticatedly architected through state-of-the-art Large Language Models (LLMs) and cutting-edge affective computing technologies, engineered to act as highly hyper-personalized conversational interlocutors that simulate nuanced anthropomorphic intimacy and retain consistent contextual memory, all to adeptly fulfill the fundamental and innate human psychological need for social relatedness (Wang et al., 2024). On the other hand, they concomitantly carry a notable risk of fostering maladaptive psychological attachment: the seamless, frictionless ease of artificial interaction can gradually desensitize users to the inevitable messiness and complexity of human relationships, ultimately rendering them increasingly intolerant of the vulnerability and imperfection that are inherent to genuine human social connections (Xie & Pentina, 2022).

In the social development domain, these AI agents also hold theoretical promise as a “safe sandbox”, a risk-free virtual space where socially isolated or anxious individuals can rehearse and hone their interpersonal communication skills, practice social initiation and response, and build interactional familiarity without the fear of real-world rejection or negative social evaluation (Croes & Antheunis, 2021). Conversely, however, over-reliance on such AI companions poses a profound social risk: they may act as a permanent, artificial substitute rather than a constructive supplement to organic, in-person community and human social bonds, thereby exacerbating social atomization and eroding users’ motivation to engage in authentic face-to-face interactions (Skjuve et al., 2021).

Most crucially, a significant and underaddressed empirical gap persists in how we evaluate human-AI relational outcomes. To date, scholarly inquiry has predominantly viewed AI companions through a functional or palliative lens, largely centering on system usability or their efficacy in mitigating negative psychological states. While existing research successfully demonstrates that AI can serve as a temporary surrogate for the isolated, far less attention has been devoted to the proactive, developmental impacts of these interactions.

To address this critical research gap, it is imperative to pivot the theoretical focus to social confidence. Grounded in social cognitive theory and self-efficacy literature, social confidence is defined as an individual’s perceived capability and belief in their power to successfully initiate social interactions, navigate complex interpersonal relationships, and execute social tasks (Bandura, 1977, 1997; Smith & Betz, 2000). Social confidence possesses distinct theoretical importance here because it functions as the pivotal psychological mediator determining the ultimate value of AI-facilitated interactions. It is the key factor that dictates whether such interactions empower users to translate the skills and comfort cultivated virtually into real-world human contexts, or instead trap them in a limiting cycle of synthetic social dependency.

Theoretical Foundation: Machine-Integrated Relational Adaptation (MIRA) Model

As AI increasingly assumes a dual function, acting simultaneously as a direct relationship partner and an indirect mediator in human social dynamics, traditional theoretical frameworks face new challenges. While established models such as the Stimulus-Organism-Response (SOR) framework offer partial explanations for these roles, they presuppose a reciprocal emotional exchange between human agents (Cheng et al., 2024). In contrast, AI occupies a unique ontological niche: it possesses the capacity for highly adaptive, personalized, and socially responsive language production, yet remains devoid of genuine mental states or lived experiences. This distinct status differentiates AI-mediated interactions from both traditional interpersonal relationships and conventional HCI.

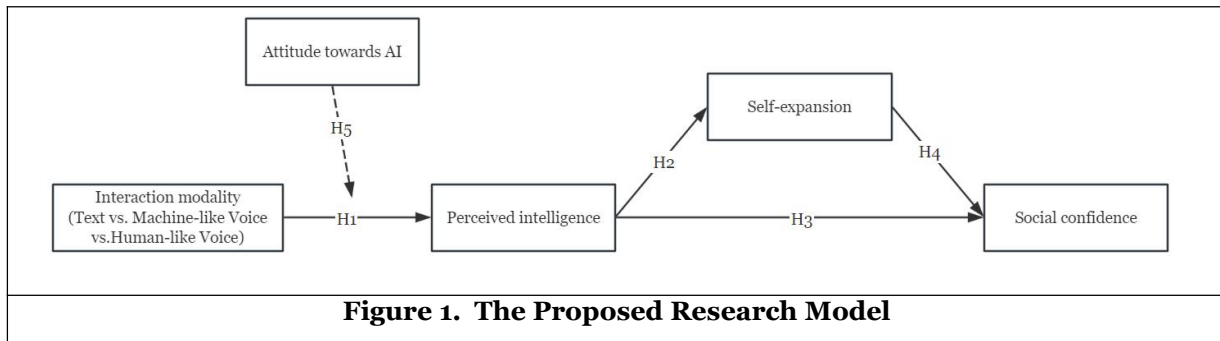
The MIRA model is proposed to address the core theoretical gap regarding how AI, as a non-conscious communication subject, becomes a meaningful relational entity in the human social ecosystem (Boyd & Markowitz, 2026). The development of this model is rooted in the integration of Attachment Theory (Ainsworth et al., 1991; Mikulincer et al., 2009), Social Exchange Theory (Blau, 1964; Homans & George, 1958) and Epistemic Trust Theory (Koenig & Harris, 2005; Sperber et al., 2010). Its core motivation stems from the observation that AI’s impact on human psychology and sociality far exceeds its functional utility; it not only simulates interpersonal interaction patterns but also reshapes the essence of relational dynamics.

Structurally, the MIRA model serves as a modular framework comprising four interrelated components: antecedent (e.g., AI system attributes such as voice modality), core mechanisms (emergent psychological processes including psychological proximity and trust), outcomes (ranging from relational enhancement to social dependence), and moderating factors (such as user characteristics) that adjust interaction strength. By adopting MIRA as our overarching theoretical lens, this study systematically examines how AI voice modality functions as a key antecedent to influence social confidence through specific cognitive and psychological mechanisms.

This study conceptualized social confidence as the primary outcome variable within the MIRA model. As a central behavioral indicator, social confidence serves as a critical bridge between transient human-AI interaction episodes and enduring interpersonal behavioral tendencies. It emerges from the model's foundational mechanisms, specifically, the dual relational role of AI as both an interactive partner and a mediating agent. Drawing on three interrelated processes, linguistic reciprocity, emotional proximity, and interpersonal trust, AI affords low-stakes, scaffolded social practice that strengthens users' expressive efficacy and situational social confidence. Over repeated interactions, this context-specific confidence progressively generalizes to human-human settings, thereby fostering adaptive social functioning and initiating a self-reinforcing developmental cycle.

Hypothesis Development

Based on the MIRA theory, we have constructed a research model. The specific variables and influencing paths of the model are shown in the Figure 1.



The Effect of AI Voice Modality

Within the MIRA framework, AI System Attributes are classified as critical antecedents that shape the initial context of the human-AI relationship. Specifically, high-fidelity anthropomorphic cues, most notably vocal characteristics, play a pivotal role in triggering a user's attribution of "mental ability" to the agent. In the field of Human-Robot Interaction (HRI), this attribution is conceptualized as Perceived Intelligence: a user's subjective evaluation of an agent's competence, knowledge, and judgment based on its behavioral performance (Bartneck et al., 2009; Tekerek et al., 2026). Since current AI has not achieved full autonomous consciousness, this subjective perception serves as the primary determinant of interaction quality.

A human-like voice is expected to increase the perceived intelligence of an AI companion because users evaluate intelligence not only from what an agent says, but also from how it communicates. Vocal cues such as natural prosody, appropriate pacing, and expressive intonation can signal communicative competence, adaptability, and contextual sensitivity, which are commonly associated with intelligence. This is especially important for AI companions, whose intelligence is judged in ongoing social interaction rather than solely through task performance. The voice principle further suggests that information presented in a human voice is processed more fluently than information delivered in a machine voice, because unnatural prosody requires greater cognitive effort to interpret (Mayer, 2005). When interaction feels easier to process and more socially natural, users are more likely to attribute greater capability and sophistication to the AI itself (Charles et al., 2006). Compared with text, which lacks vocal and paralinguistic cues, and machine voice, which may sound rigid or mechanical, a human-like voice more strongly conveys responsiveness, nuance, and contextual sensitivity. These cues enhance social presence and make the AI companion appear more

psychologically and communicatively capable, leading users to perceive it as more intelligent. Therefore, we propose

H1: Human-like voice will lead to higher perceived intelligence of an AI companion than text and machine voice.

The Mediating Role of Perceived Intelligence and Self-Expansion

A theoretical framework for human-AI relational dynamics explains how functional interactions evolve into meaningful connections via core psychological mechanisms. Drawing on the principle of “Psychological Proximity” and integrating it with Social Exchange Theory, we propose that when users perceive an AI agent as highly intelligent, they will categorize it as a valuable social resource (Gur et al., 2025). To operationalize this mechanism, we introduce Self-Expansion Theory (Aron & Aron, 1986). This theory asserts that individuals have a fundamental motivation to expand their self-concepts by incorporating the resources, perspectives, and identities of capable partners. In the context of human-AI interaction, an AI companion with high perceived intelligence is not just a tool but a partner that offers “cognitive expansion”. Interaction with such an agent provides users with new viewpoints, stimulates thinking, and broadens their worldview. Thus, the perception of intelligence acts as a catalyst for the psychological process of self-expansion. Therefore, we propose

H2: Users’ perceived intelligence of AI leads to higher sense of self-expansion.

The concluding phase of human-AI relational development centers on relational outcomes, with a core principle of “relational enhancement” positing that AI can serve as a “social scaffold” (Ertugruloglu et al., 2023). When users undergo self-expansion, feeling supported and cognitively enriched by the AI’s capabilities, this internal psychological satisfaction manifests as externalized confidence. We define this outcome as Social Confidence. Social confidence refers to individuals’ belief in their capability to initiate and maintain social relationships and execute social tasks successfully (Bandura, 1977).

To understand how interacting with an AI companion translates into a user’s own social confidence in complex human settings, it is necessary to bridge perceived intelligence and externalized self-efficacy. When an AI companion exhibits high perceived intelligence, often amplified by the nuanced, human-like qualities of voice modality, it ceases to be a mere tool and becomes a sophisticated conversational partner. A highly intelligent AI is capable of simulating the unpredictability, responsiveness, and depth of complex human interactions. When users successfully navigate deep, meaningful conversations with a highly intelligent AI, they undergo self-expansion. They absorb new perspectives, refine their communication strategies, and feel cognitively enriched by the AI’s capabilities. Because the AI is perceived as intelligent enough to mimic complex human dynamics, the user’s successful interaction with it constitutes a powerful “mastery experience” (Bandura, 1977). The AI’s social scaffold allows users to internalize this support and their newly expanded relational skills. By equipping users with a broader repertoire of conversational resources and a safe space to practice them, this internal psychological satisfaction manifests as externalized confidence, empowering them to tackle the complexities of human-to-human social settings.

Consequently, we propose a serial mediation model in which the voice modality triggers this chain of psychological events. The richness of the voice modality enhances the AI’s perceived intelligence, which in turn drives user self-expansion, ultimately culminating in heightened social confidence in broader human contexts. Therefore, we propose the following hypotheses:

H3: Users’ perceived intelligence of AI companion enhances users’ social confidence.

H4: Perceived intelligence and self-expansion play a serial mediating role between AI voice modality and social confidence.

The Moderating Role of Attitude towards AI

Public attitudes toward AI are complex and context-dependent (Schepman & Rodway, 2020), functioning as a cognitive filter that systematically moderates the impact of AI interaction features (e.g., human-like voice) on user acceptance. Specifically, this moderating effect operates through distinct mechanisms across different attitude groups: for users with low attitudes (skeptics or neutral individuals), a human-like voice serves as a powerful compensatory mechanism to break down psychological barriers, amplify perceived AI

intelligence, and substantially strengthen acceptance; for users with high attitudes (enthusiasts), their inherent positive predispositions already attribute high potential to AI regardless of interface, so the marginal gain of a human-like voice is diminished by a ceiling effect, weakening the feature's influence.

H5: Users' attitudes toward AI moderate the relationship between voice modality and perceived intelligence, such that the positive effect of a human-like voice on perceived intelligence is weaker for users with more positive attitudes toward AI.

Methodology

Overall Design

This study adopted a 14-day longitudinal field experiment with a one-way between-subjects design (AI Modality: Neutral Text vs. Machine-like Voice vs. Human-like Voice). A sample of 125 undergraduate students ($Mage=21.22$) was recruited from a major Asian university. The procedure comprised three stages: baseline, intervention, and evaluation. Initially, participants completed a baseline survey assessing demographics and pre-existing attitudes toward AI. Subsequently, they were randomly assigned to one of three conditions and required to engage in daily 10-minute interactions with a custom-developed AI agent for two weeks. The experimental manipulation strictly controlled the communication channel (text vs. voice) and vocal quality (machine-like vs. human-like) while maintaining consistent interaction content. To evaluate the trajectory of relational development, key constructs, Perceived Intelligence, Self-Expansion, and Social Confidence, were measured at both the midpoint (Day 7) and the conclusion (Day 14) of the experiment.

Manipulations

To examine how AI voice modality influences users' perceived intelligence and self-expansion, we manipulated the AI agent's communication interface across three experimental conditions. In the neutral text control condition, participants interacted with a text-only chatbot, with all AI responses presented via text bubbles and no auditory output. In the machine-like voice condition, participants received audio responses from a conventional text-to-speech system that featured a flat tone, a steady rhythm, and minimal emotional inflection to convey a robotic impression. In the human-like voice condition, the AI employed an advanced neural text-to-speech system with natural prosody, appropriate pauses, and emotional cues, such as warmth and empathy to simulate human-like speech. To isolate the effect of presentation modality, the underlying dialogue generation model was held constant across all three groups, ensuring that any differences in outcomes could be attributed to modality rather than conversational content.

Measurement

In this study, we examined users' perceptions of the AI agent and their psychological states across the modalities. Items measuring Perceived Intelligence were adapted from Bartneck et al. (2009) and Tekerek et al. (2026), to assess users' subjective evaluations of the AI's competence and judgment. Self-Expansion was measured using items adapted from Aron(1986) that captured the degree to which the interaction expanded users' perspectives and sense of self. Social Confidence was assessed based on Bandura's (1977) framework, evaluating users' confidence in social situations following the intervention. We further posit that the effect of voice modality is contingent on users' pre-existing Attitude towards AI, measured using the General Attitudes towards Artificial Intelligence Scale (GAAIS). All items were rated on a 7-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree).

Preliminary Results

The study commenced with a rigorous analysis of the collected data to verify the effectiveness of the experimental manipulations. First, we performed Levene's test for equality of variances across the three experimental groups (Neutral Text vs. Machine-like Voice vs. Human-like Voice). Preliminary checks confirmed the homogeneity of variances across all variables ($p>.05$). Results also revealed a significantly higher mean anthropomorphism rating in the human-voice condition ($M = 3.800$) compared to the text-

based condition ($M = 3.122$, $p = 0.027$). This significant difference confirms that the two conditions successfully elicited distinct levels of perceived humanness, thereby validating the efficacy of our experimental manipulation.

Subsequently, a one-way ANOVA was conducted to examine group differences. The results indicated no significant differences in initial Attitude towards AI among the groups ($F(2, 122) = 2.42$, $p = .093$), suggesting that baseline attitudes were consistent across conditions and randomization was successful.

Regarding the mediating variable, the linear contrast was significant, indicating that perceived intelligence increased with the level of anthropomorphism. Specifically, LSD post-hoc tests revealed that the Human-like Voice group reported significantly higher perceived intelligence ($M=5.26$, $SD=.73$) compared to the Neutral Text group ($M=4.81$, $SD=1.02$; $p=.032$), partially supporting our prediction regarding the impact of voice modality. For Self-Expansion ($F(2,122)=1.27$, $p=.285$) and Social Confidence ($F(2,122)=0.37$, $p=.695$), the omnibus ANOVA did not detect significant differences solely based on group assignment, which necessitates the further examination of the proposed mediation and moderation mechanisms.

Providing initial support for H1, ANOVA, and post-hoc LSD tests revealed that users in the human-like voice condition perceived the AI as significantly more intelligent ($M=5.26$) compared to those in the neutral text condition ($M=4.81$, $p=.032$). However, simple group comparisons did not yield significant direct effects for self-expansion (H2) or social confidence (H3). This absence of direct main effects suggests that the influence of AI modality is likely indirect and conditional rather than immediate. Consequently, to fully validate the proposed serial mediation (H4) and the moderating role of attitude (H5), a more comprehensive moderated mediation analysis was conducted and the results generally supported our hypotheses.

Conclusion and Expected Contributions

This study takes AI voice modality as the research starting point and based on the MIRA model, explores its influencing mechanisms and boundary conditions on users' social confidence. It has made innovative research contributions in theoretical expansion, psychological mechanism excavation, and practical subdivision, providing new ideas and a basis for theoretical research and the design application of AI multimodal interaction.

First, this study advances the theoretical boundaries of the MIRA model in the context of AI multimodal interaction. This study enriches MIRA by explicitly validating its proactive "relational enhancement" pathway through a sensory lens. By defining voice modality as the core sensory antecedent, incorporating perceived intelligence and self-expansion as the emergent psychological mechanisms, and positioning real-world social confidence as the ultimate relational outcome, we supplement MIRA's operational pathways. This research design empirically verifies that non-textual, sensory AI cues do not merely foster relational dependency; rather, they serve as a developmental catalyst that actively reshapes users' real-world psychological resources. Furthermore, introducing users' attitudes toward AI as a moderating factor refines the boundary conditions of MIRA, clarifying how individual predispositions alter the strength of human-AI relational adaptation.

Second, this study supplements the Self-Expansion theory by demonstrating that humans can undergo meaningful self-expansion by interacting with a non-conscious, artificial entity. By revealing the internal logic whereby a human-like voice enhances perceived intelligence, which subsequently drives self-expansion, our findings prove that users can cognitively internalize a machine's simulated conversational capabilities. This fundamentally expands the ontological scope of Self-Expansion Theory: it establishes that a highly intelligent-sounding AI can serve as a legitimate, resource-rich "Other" that users include in their self-concept. Consequently, this study maps exactly how artificial psychological resources can be transformed into authentic, real-world social confidence, enriches the theoretical research on the activation and improvement of users' psychological resources in the field of HCI.

Third, this study identifies that users' attitudes toward AI are important boundary conditions for the functioning of AI voice modality, providing a key practical basis for the differentiated design and precise implementation of AI interaction systems. The research results provide a precise design direction for AI product developers: formulating differentiated voice interaction strategies for user groups with different attitudes toward AI. In particular, for users with low to moderate attitudes who hold a reserved attitude

toward AI, high-fidelity human-like voice design can be used to break their psychological barriers and improve their perception and acceptance of AI; whereas, for users with high attitudes, the focus can be on the optimization of voice functions rather than the anthropomorphic upgrade of forms.

Despite these contributions, several notable limitations must be acknowledged. A primary constraint pertains to the study's generalizability, which is inherently limited by both the sample size and its demographic composition. Our participants were predominantly young undergraduate students, a cohort characterized by specific social developmental needs and a higher baseline of digital literacy. To address these limitations, the next phase of our research will implement a more robust test of the moderated mediation framework to rigorously verify the mediating roles of perceived intelligence and self-expansion, alongside the moderating effect of users' attitudes toward AI. Furthermore, we will replicate the experiment with a larger and more diverse demographic sample to enhance external validity. To complement the quantitative data, we also plan to integrate qualitative insights from semi-structured interviews and open-ended questionnaires, which will capture the nuanced, subjective experiences users have with different AI voice modalities. Finally, building on the outcomes of this study, we aim to develop and investigate AI agents equipped with richer, multimodal interactive capabilities to further expand the scope of our research.

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